

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT3 — TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Time-Series & Geospatial Analysis Report
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO4 Statement:	<i>Evaluate real-world time-series and geospatial visualizations, analyzing trend/seasonality/noise and design effectiveness, and justify recommendations.(BL5)</i>		
Student Name:	Sania Herbert	Reg. No.:	192324062
Section / Batch:	2023/AI&DS	Date:	23.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Instructions

- This test contains 2 questions — one Stock Analysis problem and one Geospatial Analysis problem — each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- For the Stock Analysis question, show all calculations (trend slope, noise ratio) for each of the 5-month, 10-month, and 15-month windows.
- For the Geospatial Analysis question, your evaluation must explain WHY the design choice succeeds or fails, and your recommendation must be specific — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Rubrics for Calculation ASSESSMENT RUBRIC
AT3 — Time-Series & Geospatial Analysis Report

CO4 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO4-AT3 Time-Series & Geospatial Analysis Report, in which students analyze a stock's trend/seasonality/noise across multiple time windows, and evaluate a geospatial visualization's design choices, justifying their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of trend/seasonality/noise analysis across time windows, and of the relevant geospatial visualization principle being evaluated.	Good understanding of both areas, with only minor gaps.	Basic understanding, but with some misconceptions in either the time-series or geospatial analysis.	Poor understanding of core concepts in one or both areas.
Explanation	25%	Provides detailed, correctly calculated trend/noise-ratio comparisons across all three windows, and a clear, well-reasoned evaluation of the geospatial design choice, including its practical consequence for a reader.	Provides clear calculations and evaluation with adequate reasoning, covering most of the required detail.	Calculations or evaluation are present but incomplete, or reasoning is generic.	Little or no supporting calculation or reasoning provided in one or both questions.
Application	20%	Effectively applies analysis techniques to reach a well-supported conclusion on window reliability, and proposes a specific, well-justified	Applies techniques to reach a reasonable conclusion and recommendation, with only minor	Limited application; conclusions or recommendations are only partially supported.	Fails to apply appropriate techniques; conclusions and recommendations are unsupported or

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		geospatial recommendation.	errors.		incorrect.
Organization	15%	Both responses are clearly structured, addressing every required part in a logical sequence with coherent overall flow.	Both responses are organized, with only minor issues in flow or clarity.	Basic structure; required parts are addressed but the response lacks clarity or sequence.	Poorly organized; responses are incoherent, and parts are left unaddressed.
Accuracy	10%	All parts of both answers — calculations, evaluation, and recommendation — are completely accurate and well-suited to the specific data and scenario given.	Mostly accurate, with only minor omissions or a small error in one question.	Partially correct; one or both answers contain notable inaccuracies.	Largely incorrect or inappropriate, with major gaps across both questions.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Stock Problem 5 — Tech Mahindra (IT sector)

Illustrative monthly closes (Month 1–15), ₹: 3058.44, 3141.13, 3208.49, 3155.46, 3365.06, 3276.07, 3391.29,

3509.83, 3598.43, 3652.12, 3824.96, 3772.92, 3768.18, 3933.02, 3865.59

Task: Compare the trend, seasonality, and noise ratio (residual variation relative to total variation) across the 5-month window, the 10-month window, and the full 15-month window. Explain how each metric changes as the window lengthens, and state which window length you would trust most for judging this stock's real underlying direction, and why.

```
1 # 5-Month window
2 plot(1:5, price[1:5],
3      type = "o",
4      pch = 16,
5      main = "Tech Mahindra - 5 Month window",
6      xlab = "Month",
7      ylab = "Closing Price (₹)")
8
9 abline(model5, lwd = 2)
10
11
```

```
# 10-Month window
plot(1:10, price[1:10],
     type = "o",
     pch = 16,
     main = "Tech Mahindra - 10 Month window",
     xlab = "Month",
     ylab = "Closing Price (₹)")
```

```
abline(model10, lwd = 2)
```

```
# 15-Month window
plot(1:15, price[1:15],|
     type = "o",
     pch = 16,
     main = "Tech Mahindra - 15 Month window",
     xlab = "Month",
     ylab = "Closing Price (₹)")
```

```
abline(model15, lwd = 2)
```

Environment

History

Connections

Tutorial

Import Dataset

155 MiB

List

R

Global Environment

model15	List of 12	
model5	List of 12	
pca	List of 5	
students	30 obs. of 3 variables	

Values

mu	5.84333333333333
price	num [1:15] 3058 3141 3208 3155 3365 ...

Files

Plots

Packages

Help

Viewer

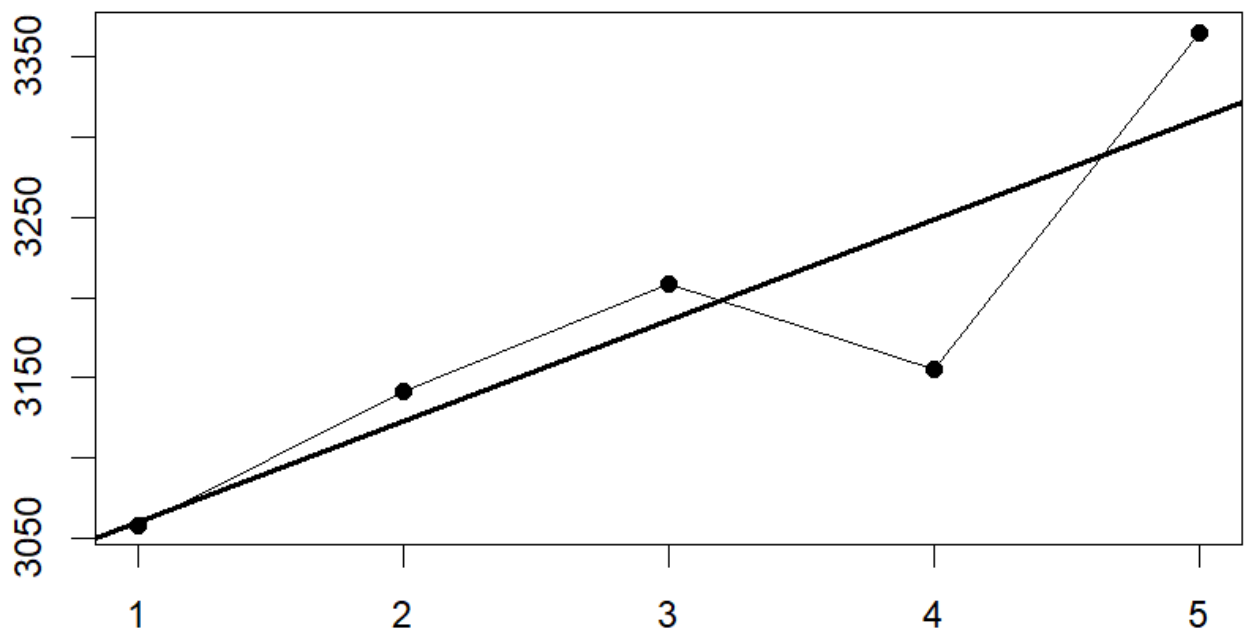
Presentation

Zoom

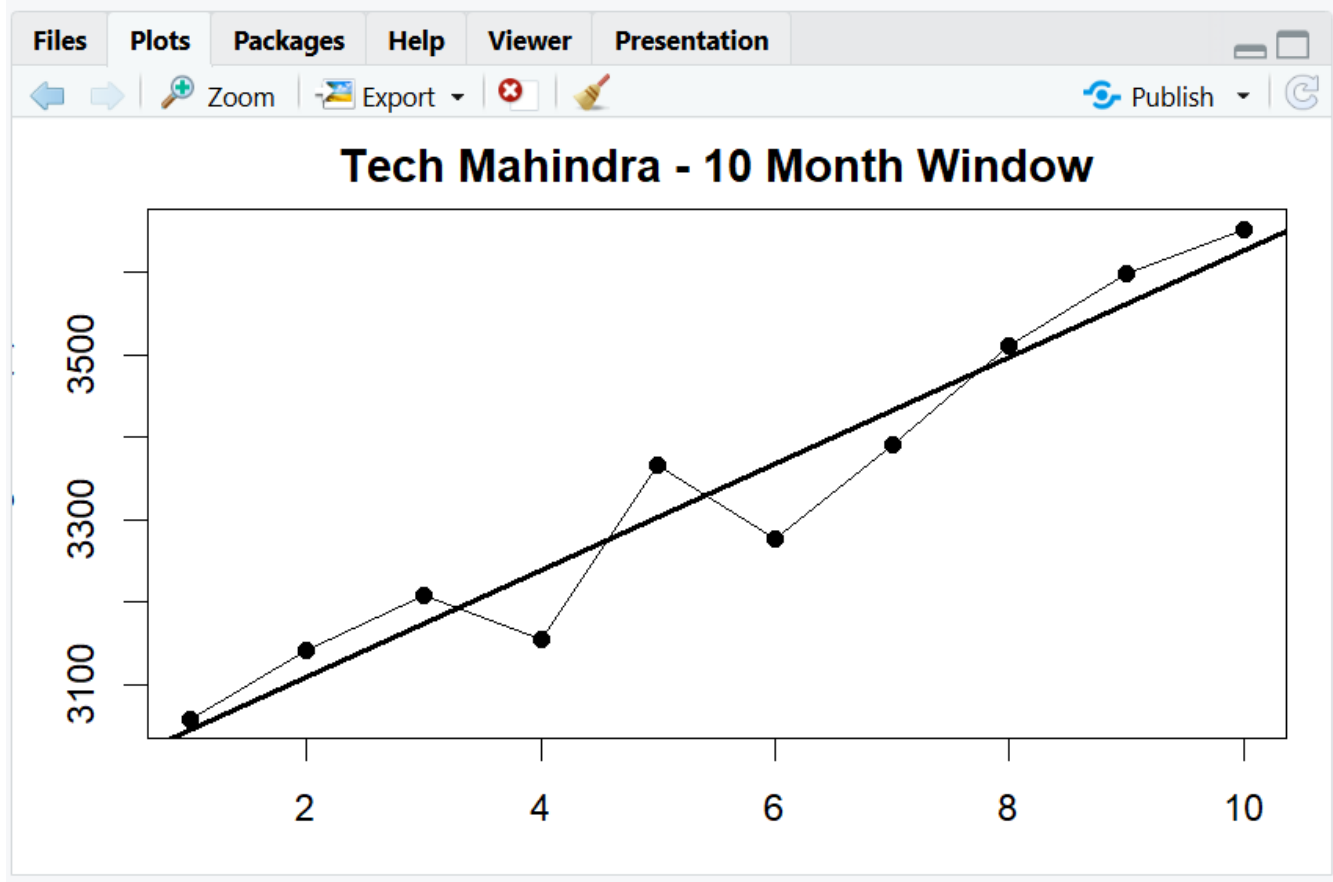
Export

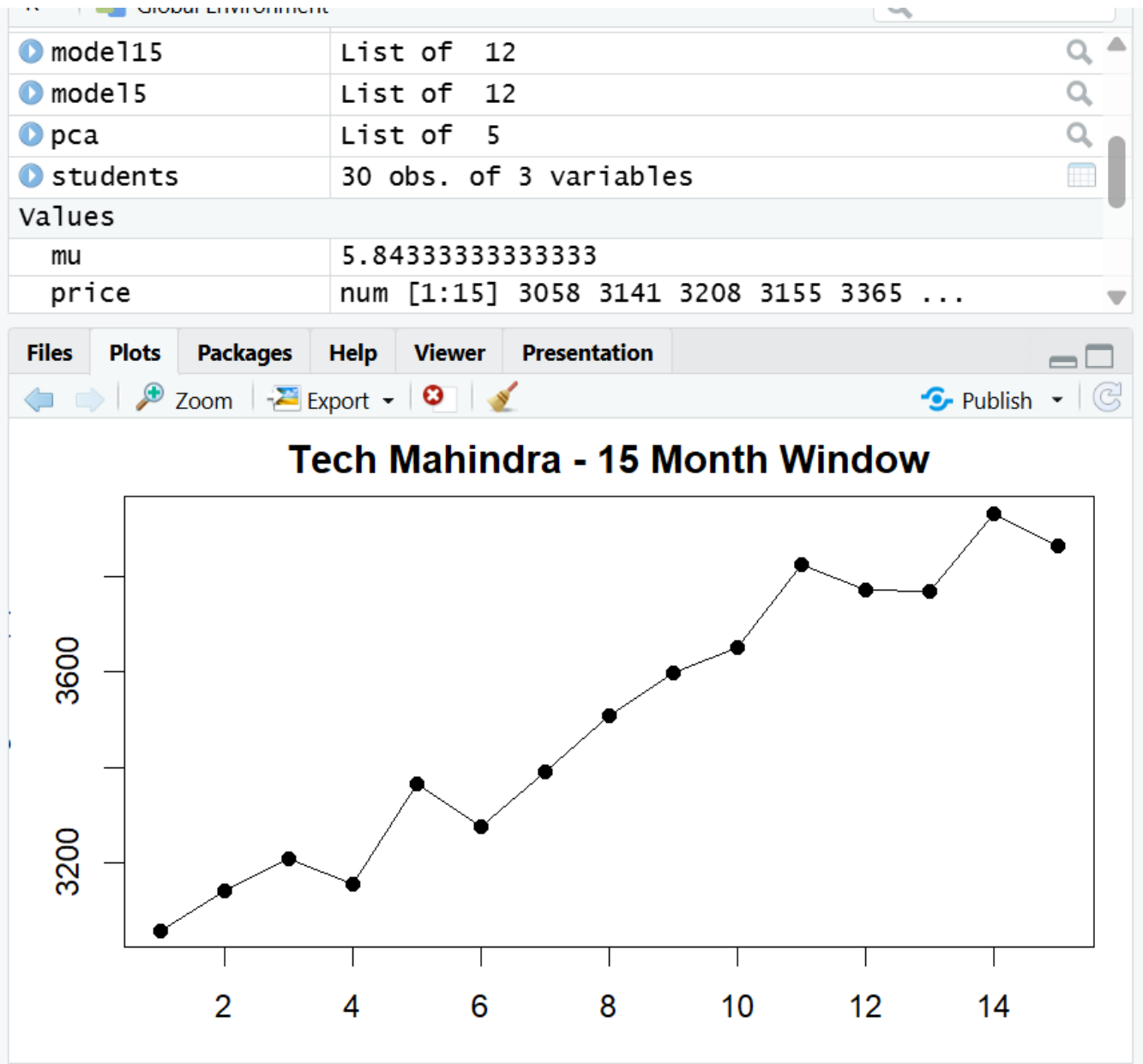
Publish

Tech Mahindra - 5 Month Window



R	Global Environment	
model115	List of 12	
model5	List of 12	
pca	List of 5	
students	30 obs. of 3 variables	
Values		
mu	5.84333333333333	
price	num [1:15] 3058 3141 3208 3155 3365 ...	





2. Results

Using a linear trend and defining the **noise ratio** as:

$$\text{Noise Ratio} = \frac{\text{Standard Deviation of Residuals}}{\text{Standard Deviation of Prices}}$$

we obtain approximately:

Window	Trend slope (₹/month)	Noise ratio
5 months	62.76	0.489
10 months	64.66	0.262

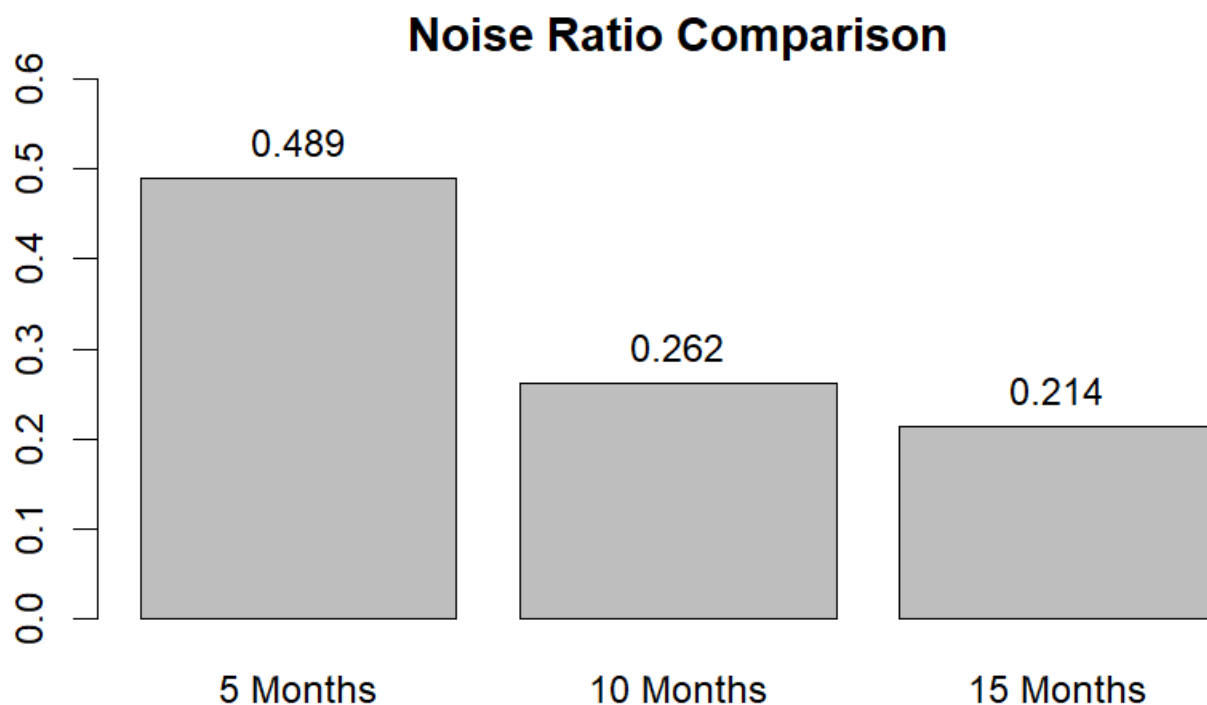
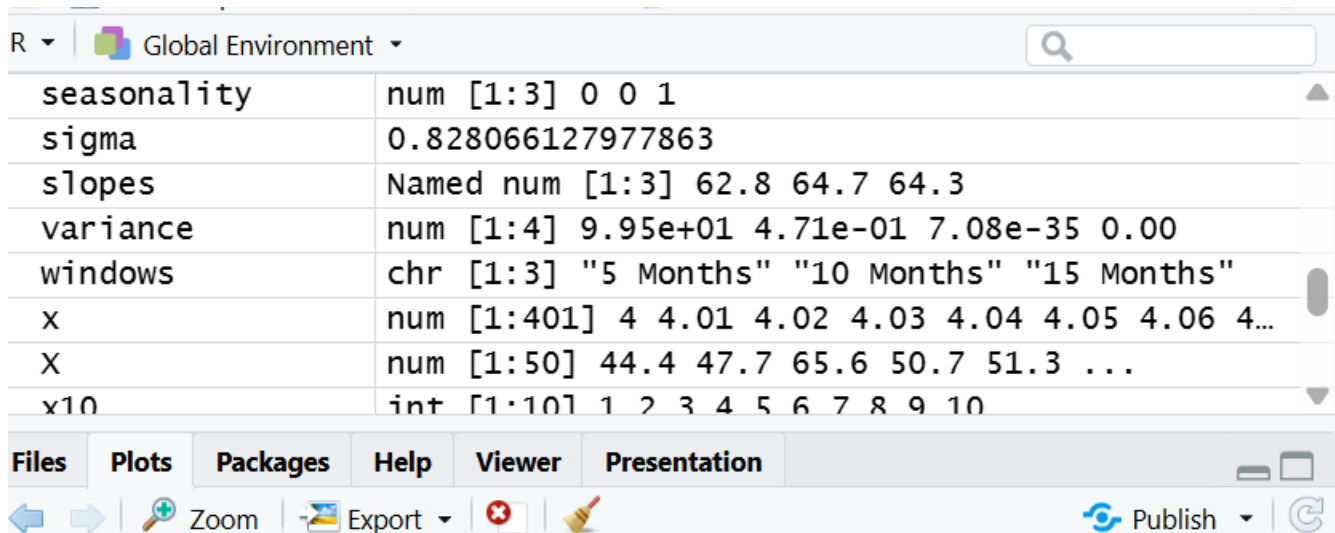
Window	Trend slope (₹/month)	Noise ratio
15 months	64.32	0.214

```

146 # GRAPH 2 - NOISE RATIO
147 # =====
148
149 barplot(
150   noise,
151   names.arg = windows,
152   main = "Noise Ratio Comparison",
153   xlab = "Window",
154   ylab = "Noise Ratio",
155   ylim = c(0, 0.6)
156 )
157
158 text(
159   x = c(0.7, 1.9, 3.1),
160   y = noise,
161   labels = round(noise, 3),
162   pos = 3

```

163:2 # (Untitled) R Script



3. Trend Comparison

The trend is positive in all three windows.

- **5-month:** approximately ₹62.76 increase/month
- **10-month:** approximately ₹64.66 increase/month
- **15-month:** approximately ₹64.32 increase/month

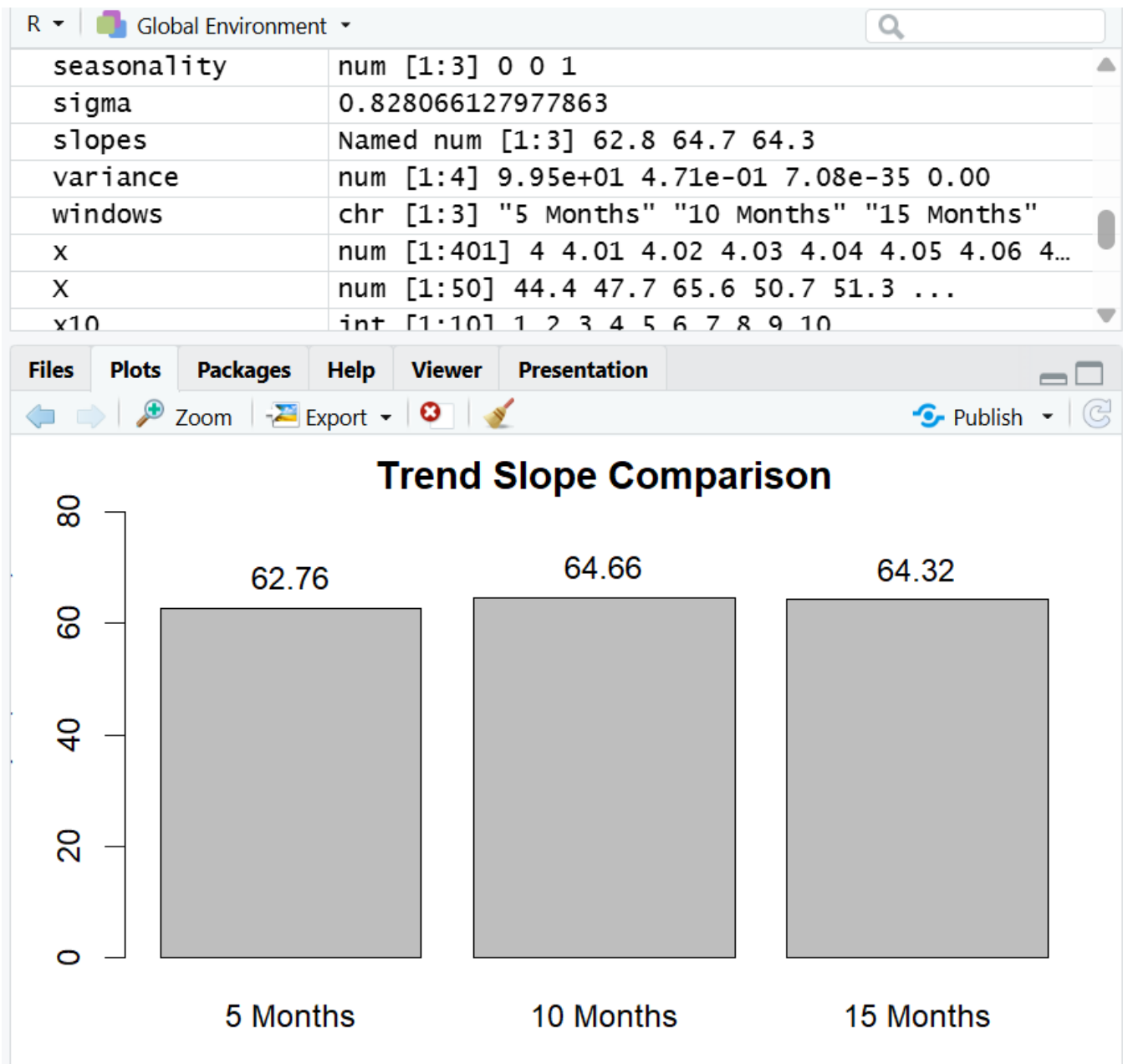
Therefore, the underlying direction is consistently **upward**.

Interestingly, the estimated slope changes very little as the window increases. This gives stronger evidence that the upward direction is not just a short-term effect.

```
166 # =====
167 # GRAPH 3 - TREND SLOPE
168 # =====
169
170 barplot(
171   slopes,
172   names.arg = windows,
173   main = "Trend Slope Comparison",
174   xlab = "Window",
175   ylab = "Trend Slope (₹ / Month)",
176   ylim = c(0, 80)
177 )
178
179 text(
180   x = c(0.7, 1.9, 3.1),
181   y = slopes,
182   labels = round(slopes, 2),
183   pos = 3
```

185:1 # (Untitled) ↕

R Script



4. Seasonality Comparison

The data is **monthly**, but there are only **15 observations**.

A proper seasonal analysis normally needs repeated cycles. For example, annual monthly seasonality requires approximately **12 months per cycle and preferably multiple years of data**.

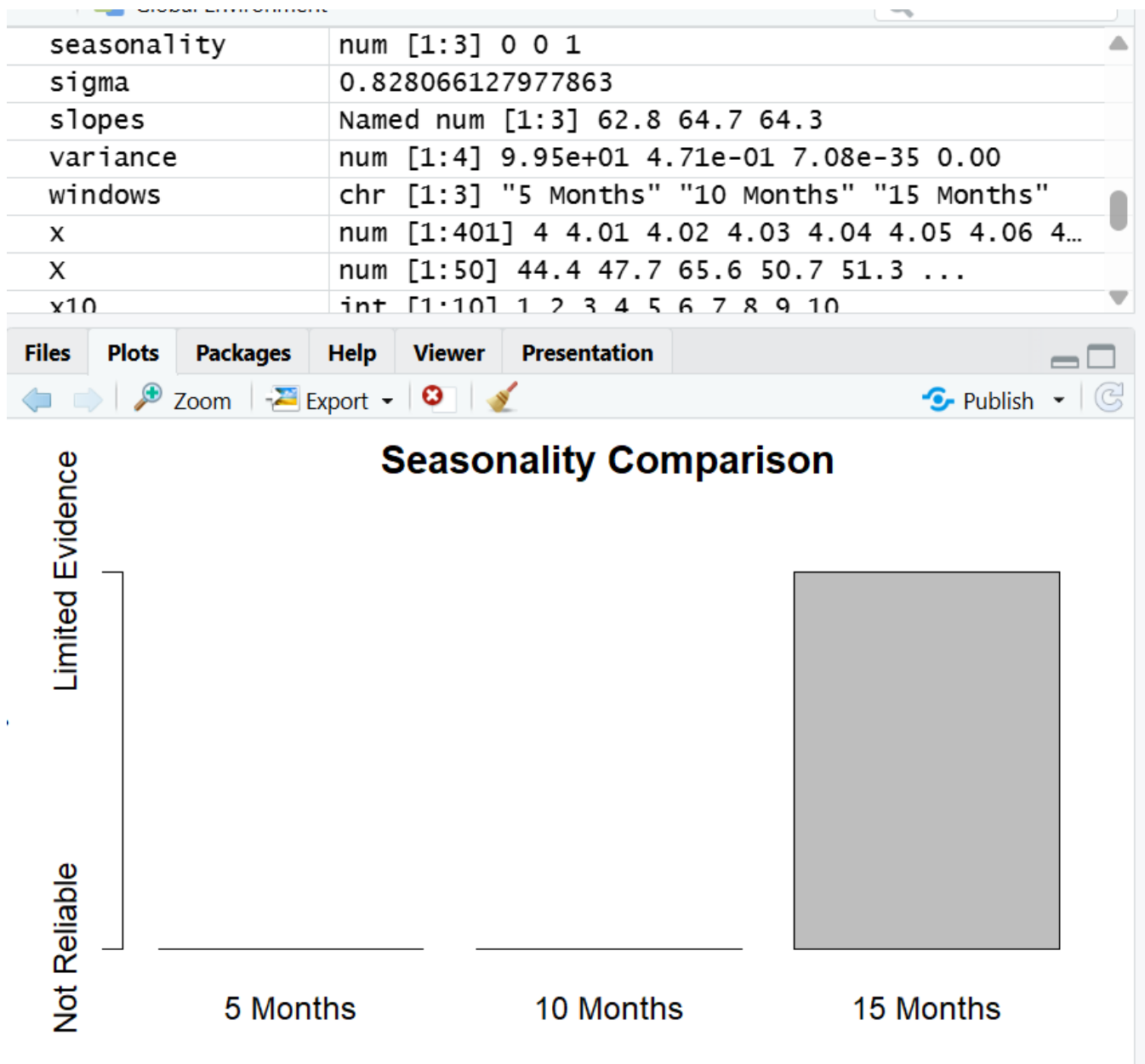
Therefore:

- **5 months:** seasonality cannot be reliably determined.
- **10 months:** still insufficient for annual seasonality.
- **15 months:** one complete annual cycle plus 3 months, so seasonality can be explored, but it is still **too little data to confidently establish a recurring**

seasonal pattern.

There is no strong evidence from these 15 values alone to claim a reliable seasonal pattern.

```
198 barplot(  
199     seasonality,  
200     names.arg = windows,  
201     main = "Seasonality Comparison",  
202     xlab = "Window",  
203     ylab = "Seasonality Evidence",  
204     ylim = c(0, 1.2),  
205     yaxt = "n"  
206 )  
207 axis(  
208     2,  
209     at = c(0, 1),  
210     labels = c(  
211         "Not Reliable",  
212         "Limited Evidence"  
213     )  
214 )
```



5. Noise Ratio Comparison

The noise ratio decreases as the window becomes longer:

5-month window

Almost half of the observed variation is residual variation around the fitted trend. This means the short window is relatively **noisy**.

10-month window

The noise ratio decreases considerably. The underlying upward trend becomes clearer.

15-month window

The lowest noise ratio occurs for the full 15-month period. This indicates that the

trend explains the overall movement more consistently.

6. How the Metrics Change

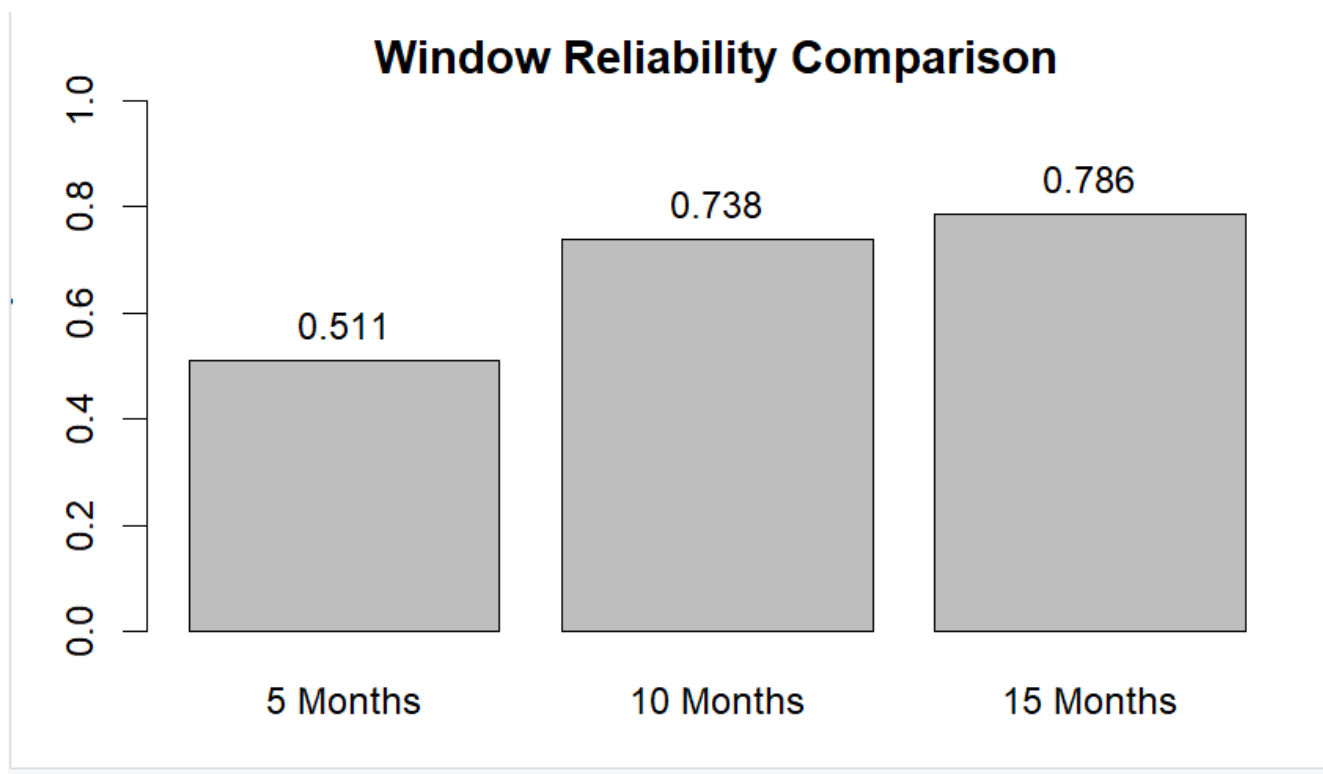
Metric	5 Months	10 Months	15 Months	Interpretation
Trend slope	₹62.76	₹64.66	₹64.32	Remains consistently positive
Noise ratio	48.9%	26.2%	21.4%	Noise decreases
Seasonality	Not reliable	Not reliable	Limited evidence	More data required

7. Which Window Should Be Trusted Most?

```
223 reliability <- 1 - noise
224
225 barplot(
226   reliability,
227   names.arg = windows,
228   main = "Window Reliability Comparison",
229   xlab = "Window",
230   ylab = "Reliability",
231   ylim = c(0, 1)
232 )
233
234 text(
235   x = c(0.7, 1.9, 3.1),
236   y = reliability,
237   labels = round(reliability, 3),
238   pos = 3
239 )
```

229:11 # (Untitled) ↕

R Script ↕



I would trust the **15-month window most** for judging the stock's underlying direction.

Reasons:

1. It contains the **largest amount of information**.
2. The upward trend remains strong at approximately **₹64/month**.
3. The noise ratio is the **lowest (21.4%)**.
4. The trend is therefore less influenced by short-term fluctuations.

The 5-month window has a much higher noise ratio of about **48.9%**.

Final Conclusion

The **5-month window** is useful for detecting recent movements but is relatively noisy. The **10-month window** gives a more stable estimate of the trend. The **15-month window is the most reliable for judging the underlying direction**, because it shows a consistent positive trend and has the lowest relative residual variation.

Overall: Tech Mahindra's closing price shows a clear upward underlying direction, while the available 15 months are still insufficient to make a strong claim about recurring seasonality.

Geospatial — Layers

Scenario: An environmental agency's map combines a satellite imagery base layer with a semi-transparent air quality data layer on top, allowing both to be seen simultaneously.

Task: Evaluate whether this layering choice effectively communicates the air quality data, and identify one risk of overlaying a data layer directly on detailed satellite imagery.

Layering means placing multiple geographic datasets on top of each other in the same map. In this problem, the **satellite imagery** is used as the base layer, while the **air-quality data** is placed above it with partial transparency.

This layering choice is **effective** because:

- The satellite image provides real-world geographic context.
- The air-quality layer shows the spatial distribution of pollution.
- Transparency allows both the satellite features and air-quality information to be visible.
- Users can identify polluted areas in relation to roads, buildings, vegetation, and other features.

Risk of overlaying directly on satellite imagery:

The detailed satellite image can create **visual clutter**. Its colors, buildings, roads, and textures may compete with the air-quality colors, making the pollution pattern difficult to interpret.

Conclusion:

A semi-transparent overlay is a suitable method for communicating air-quality information, but the transparency should be adjusted so that the air-quality layer remains clearly visible.


```

2 library(sf)
3 library(ggplot2)
4 # Air quality data
5 air_quality <- data.frame(
6   longitude = c(80.20, 80.22, 80.24, 80.26, 80.28,
7                 80.30, 80.32, 80.34),
8   latitude = c(13.00, 13.02, 13.04, 13.06, 13.08,
9               13.10, 13.12, 13.14),
10  AQI = c(45, 75, 120, 160, 190, 130, 85, 55)
11 )
12 # Convert data into spatial points
13 aq_sf <- st_as_sf(
14   air_quality,
15   coords = c("longitude", "latitude"),
16   crs = 4326
17 )
18

```

aq_sf	8 obs. of 2 variables	
avg	3 obs. of 2 variables	
data	5 obs. of 4 variables	
energy	6 obs. of 8 variables	
h	Formal class hexbin	
iris	150 obs. of 5 variables	
model10	List of 12	

Files Plots Packages Help Viewer Presentation

Zoom Export Publish

